Netflix Movie Analytics: Understanding Genre Trends and Audience Preferences

1. Project Overview:

This project analyzes movie trends, audience engagement, and genre performance using Python-based exploratory analysis. Finding meaningful relationships between ratings, popularity, and content distribution, providing actionable recommendations for content investment decisions.

2. Dataset Summary:

Rows: 9837
Column : 9
Key Features: Release_Date (Movie release date)
Title (Movie title)
Popularity (Audience engagement score)
Vote_Count (Number of audience votes)
Vote_Average (Average movie rating)
Genre (Movie category)
Missing data: number of missing records per column ranged between 9-11 values
Duplicate Data: 8 duplicate records were identified and removed

3. Exploratory Data Analysis using Python:

Data Loading : Dataset was imported using Pandas

Initial Exploration:
Used `df.info()` to understand dataset structure
df.describe() to check data types and summary
df.isnull().sum() to detect missing values

]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9837 entries, 0 to 9836
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Release_Date       9837 non-null   object 
 1   Title              9828 non-null   object 
 2   Overview           9828 non-null   object 
 3   Popularity         9827 non-null   float64
 4   Vote_Count         9827 non-null   object 
 5   Vote_Average       9827 non-null   object 
 6   Original_Language  9827 non-null   object 
 7   Genre              9826 non-null   object 
 8   Poster_Url         9826 non-null   object 
dtypes: float64(1), object(8)
memory usage: 691.8+ KB
```

Missing Data Handling:
missing values represented a very small proportion of the dataset affected rows were removed to maintain data quality

Data Type Corrections:
Changes Performed:
`Release_Date` converted to datetime
`Vote_Count` converted to float
`Vote_Average` converted to float

Proper data types improve analytical accuracy and efficiency

Feature Engineering:

Release Year Extraction: release date was transformed into release year for temporal trend analysis

Rating Categorization:
Vote_Average was categorized into four rating groups to simplify interpretation
Categories:

Low Rated

Average Rated

Well Rated

Highly Rated

Genre Normalization:

Movies belonging to multiple genres were split into separate rows to improve genre level analysis

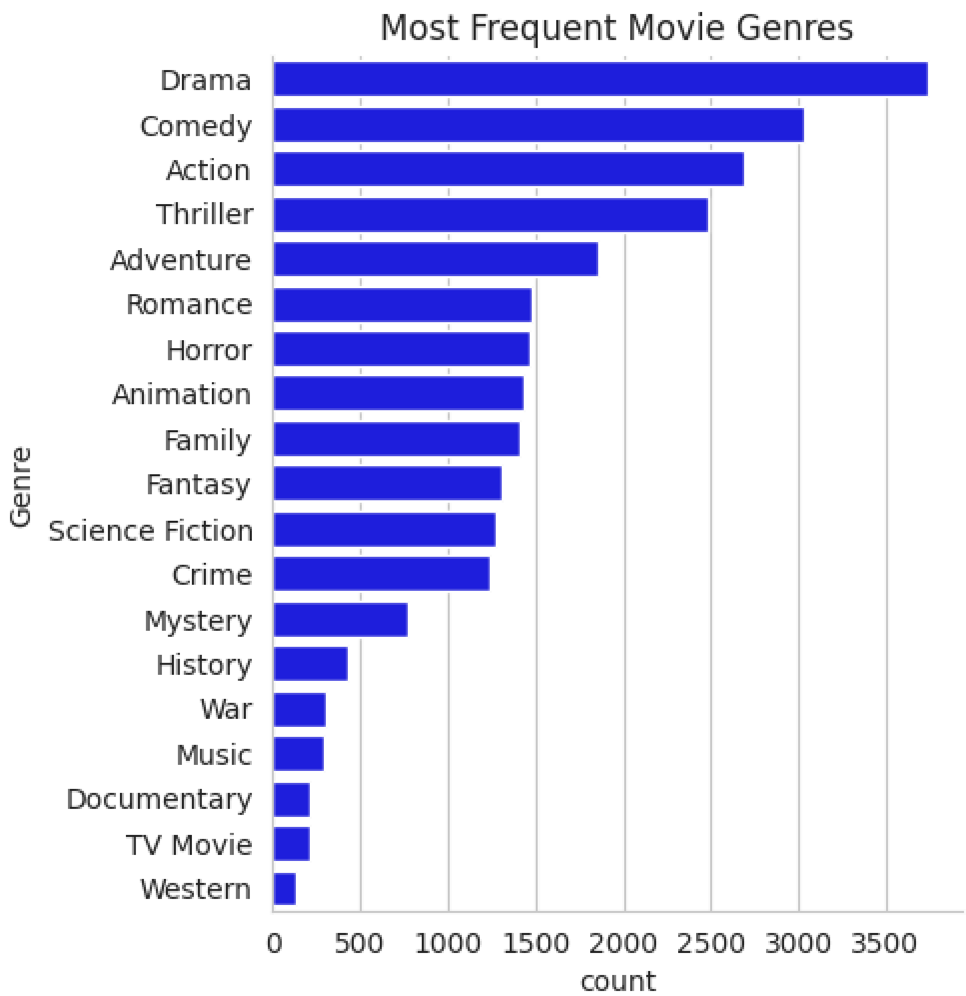
4. Data Analysis & Insights

1.What is the most frequent movie genre?

```
]: df['Genre'].describe()
```

```
]: count      25792
   unique        19
   top        Drama
   freq        3744
   Name: Genre, dtype: object
```

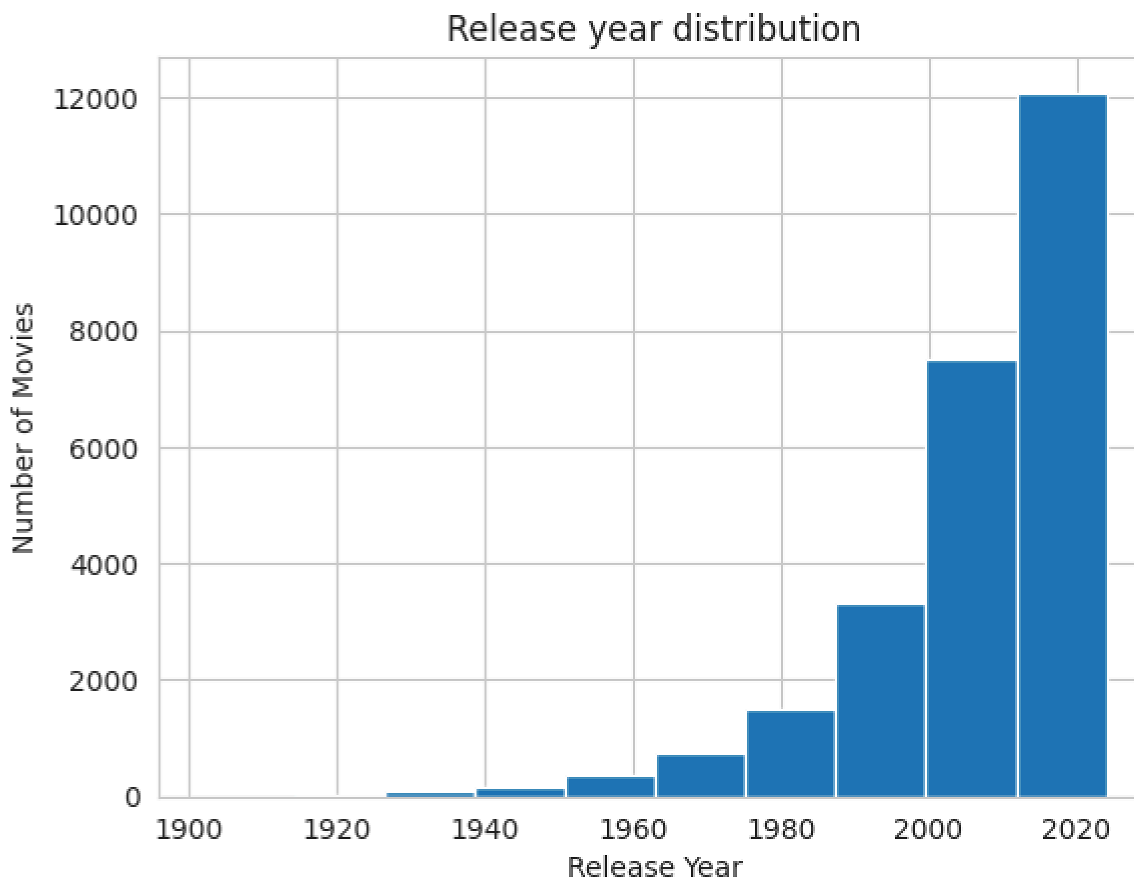
```
]: sns.catplot(y='Genre', data=df, kind='count',
               order=df['Genre'].value_counts().index,
               color='Blue')
   plt.title('Most Frequent Movie Genres')
   plt.show()
```



Drama genre is the most frequent genre in dataset and has appeared more than 14% of the times among 19 other genres

2.How are movie releases distributed across years?

```
plt.show()
```



Movie releases increased significantly after 2000

Year 2020 has the highest filming rate in dataset

3.Which movie recorded the highest popularity?

```
df[df['Popularity']==df['Popularity'].max()]
```

	Release_Year	Title	Popularity	Vote_Count	Vote_Average	Original_Language	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940.0	Highly Rated	en	Action
1	2021	Spider-Man: No Way Home	5083.954	8940.0	Highly Rated	en	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940.0	Highly Rated	en	Science Fiction

Spider-Man: No way home has the highest popularity and has genres Action, Adventure and Science fiction

4.Which movie recorded the lowest popularity?

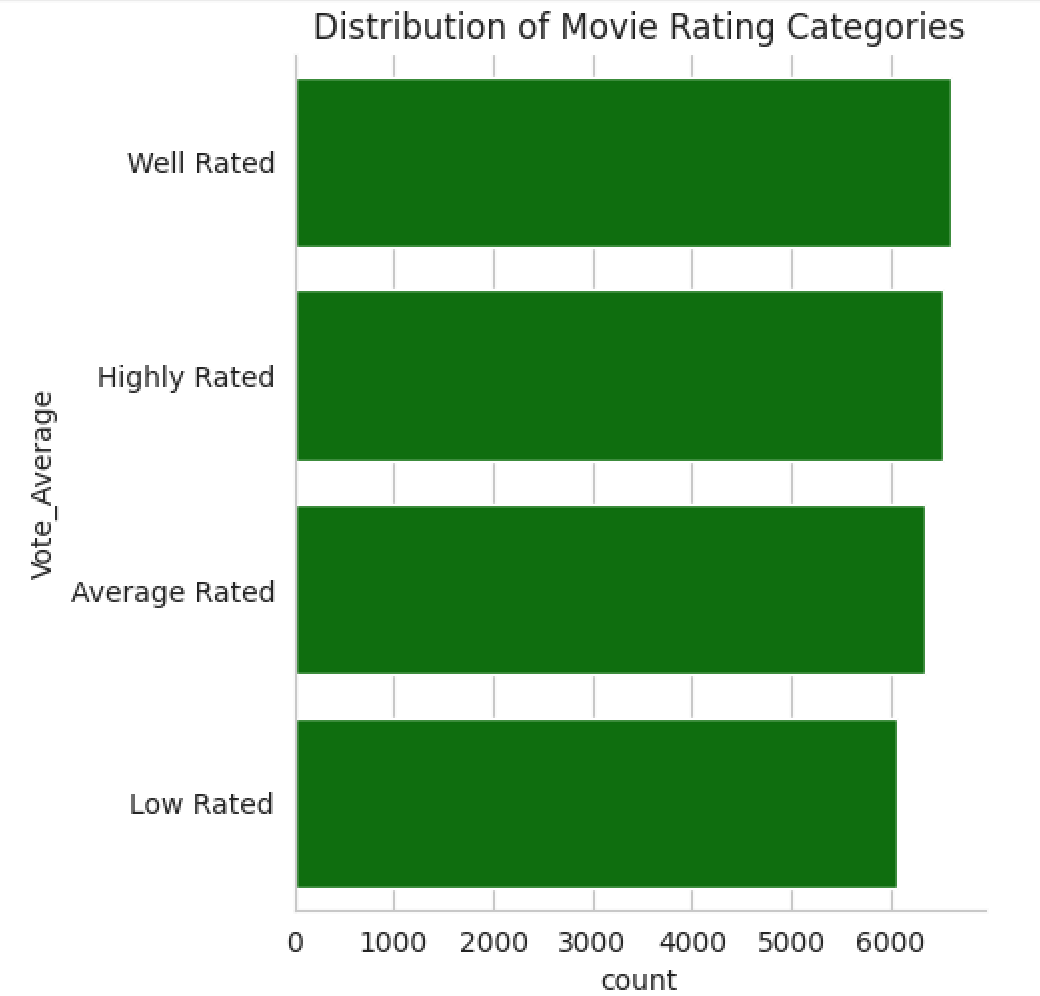
```
df[df['Popularity']==df['Popularity'].min()]
```

	Release_Year	Title	Popularity	Vote_Count	Vote_Average	Original_Language	Genre
25545	2021.0	The United States vs. Billie Holiday	13.354	152.0	Well Rated	en	Music
25546	2021.0	The United States vs. Billie Holiday	13.354	152.0	Well Rated	en	Drama
25547	2021.0	The United States vs. Billie Holiday	13.354	152.0	Well Rated	en	History
25548	1984.0	Threads	13.354	186.0	Highly Rated	en	War
25549	1984.0	Threads	13.354	186.0	Highly Rated	en	Drama
25550	1984.0	Threads	13.354	186.0	Highly Rated	en	Science Fiction

The United States vs. Billie Holiday , Threads has the lowest popularity and it has

Genre Music, Drama, History, War, Sci fi

5. Which rating category is most common?



We have 25.87% of dataset with “Well Rated” Category (6612)

5.Business Recommendations

Invest more in high-performing genres: high popularity and strong audience ratings

Monitor changing genre trends : Audience preferences evolve over time

Use audience ratings alongside popularity : Popularity alone may not indicate quality , Highly rated niche content may offer growth opportunities